Rout Dutta

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saathi

Project: ML → what does this mean?

CBIR → content based image retrieval (1)

Segmentation using clustering. → what does this mean?

(1) CBIR → locate samples from a database that are akin to a query, relying on content embedded within image.

⇒ * Strategy: calculating similarity b/w "compact vectors" by encoding both query & database images as global descriptors

* CLIP, UNICOM, CNN, GEM, Swin → 1 semantic representation per image.
[512-D, 768-D, 1024-D].

* what is "content"? → "sub regions"

content ⇒ high level semantics extracted from the backbone model.

what it includes:-
1. which objects are present?
2. their visual patterns.
3. overall style.
4. color distributions.
5. texture
6. category cues.
7. shape structure.
8. scene type.
9. Relationship b/w subregions.

→ compressed learned representation

"ONLY latent features"

* what do these vectors contain?
1) object identity
2) structural & id / color attributes
3) global composition.
(4) High level semantics
(5) Implicit visual details.

DO NOT contain "bounding boxes" / "per pixel patterns".

Q) How is a "png" image converted into a "global descriptor" vector?

Full pipeline: → gives out a set of ~~single~~ vectors $g \in R^d$.

STEP I Feed image into a backbone [CNN (1) or ViT (2)]. sizes: (one only) {512, 768, 1024}.

(i). CNN → ResNet 101 → produce feature maps.
(ii) ViT → Patch embeddings → transformer outputs.
(iii) CLIP → image encoder → final pooled embedding.

STEP II Aggregate the cluster of vectors into 1 vector per image.

Once we get a set of vectors using 2 types:-

1. CNN → ResNet / R-GeM {feature map $\in R^{C \times H \times W}$}
2. ViT/CLIP/UNICOM {vision transformers} → {patch embeddings} ↓ {transformers} ↓ {output tokens}

Step 2

Then aggregation happens:

CASE 1 CNN $\xrightarrow{\text{step 1}} R^{C \times H \times W}$

Image: $(P_1 \times P_2 \times 3)$.
C → channel [2048 channels].
H → 7
W → 7 } (1) image separated into 7×7 = 49 "regions" / "sub regions"

(2) Each position $\in R^C$ dimensional vector representation.

(3) 49 vectors for each 2048 size rep vectors.

$F_{:,h,w} \in R^C \longrightarrow \{v_1, v_2, \dots, v_{HW}\}^D$

Aggregation for [1] → (1) average pool: $g_c = \frac{1}{HW} \sum_{h,w} F_{c,h,w}$.

[2] Max pool: $g_c = \max_{h,w} \{F_{c,h,w}\}$.

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→ These are pre trained models

∴ 1) NO learning.
2) NO finetuning.
3) NO weight updates.

~~Q) What is a "QUERY"?~~

→ Single vector representation of 1 img

CASE [2]: Vision Transformers.

1) Image is cut into 16×16 / 14×14 patches.
2) 16×16 = N ⟶ "sub-images" for [EACH] img sample.
3) Each of these "N" patches are fed into a Vision transformer: $\{\hat{x}^D_N\} \in R^{D \times N}$ Transformer

$\{x^D_{CLS}\}$ → CLS token

~~(classification token) $\in R\{x^D_1, x^D_2, \ldots x^D_N\}$~~

Each patch embedding $\{\hat{x}_D\} \in R^D$.
Total number → $N = 16 \times 16$.

~~for the total of images in the entire dataset.~~

(4) $x_{CLS} \in R^D$: prepend it into vector array i.e $\{x^D_{CLS}, x^D_1, x^D_2, \ldots, x^D_N\}$.

(5) $x^{(L)}_{CLS} = f(x_{CLS}, x_1, x_2, \ldots x_N)$.
→ [summarizing vector] after 'L' layers of transformer

(6) CLIP, UNICOM ⟶ x^D_{CLS} of size D for 1 image.

(7) Aggregation NOT NECESSARY. x_{CLS} is SMART.

Now for a Data set of size Δ we have Δ representative vectors for each image

STEP [III]. we have descriptor set for each constituent image within leaf nodes.

NOTE: What the heck is a QUERY?
A) It is an image that asks: "Is this tree/car/animal in an images of your dataset", "can you tell which one?", "which one do you think matches the closest description?".

Next step: Generate query descriptor singularly using STEP [I] & STEP [II].

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~~Now are~~

STEP IV :

Use the global descriptor of QUERY (v_q), & calculate similarity (s_i) with leaf nodes. Each leaf node of each image gives some similarity.

similarity distance metric / cosine } dataset size.

$v_q \rightarrow \{ v_1, v_2, v_3, \ldots v_i \ldots v_N \}$.

N leaves — dataset size.

$\Rightarrow \{ s_1, s_2, s_3 \ldots s_i, \ldots s_N \}$

scalars

~~Speed - Accuracy tradeoff.~~

STEP IV. : OFFLINE stage. (N → total dataset)

→ Now we have to understand that we start with OFFLINE stage.

→ For CNN path ⟶ aggregated info rich vector.
For ViT path ⟶ CLS vector.

→ Now we have N such vectors.

→ Implement HKM → Hierarchical K-means. AND build search TREE.

→ Root node ⟶ K children. $\{\mu_k\}$ centroids.

→ Each i^{th} node has images ($\mu_i^k > (N)$). which are closest to the i^{th} centroid.

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Hyperparams: 1. K ⇒ K-means
↳ branching factor (AKA).
2. l ⇒ tree depth.

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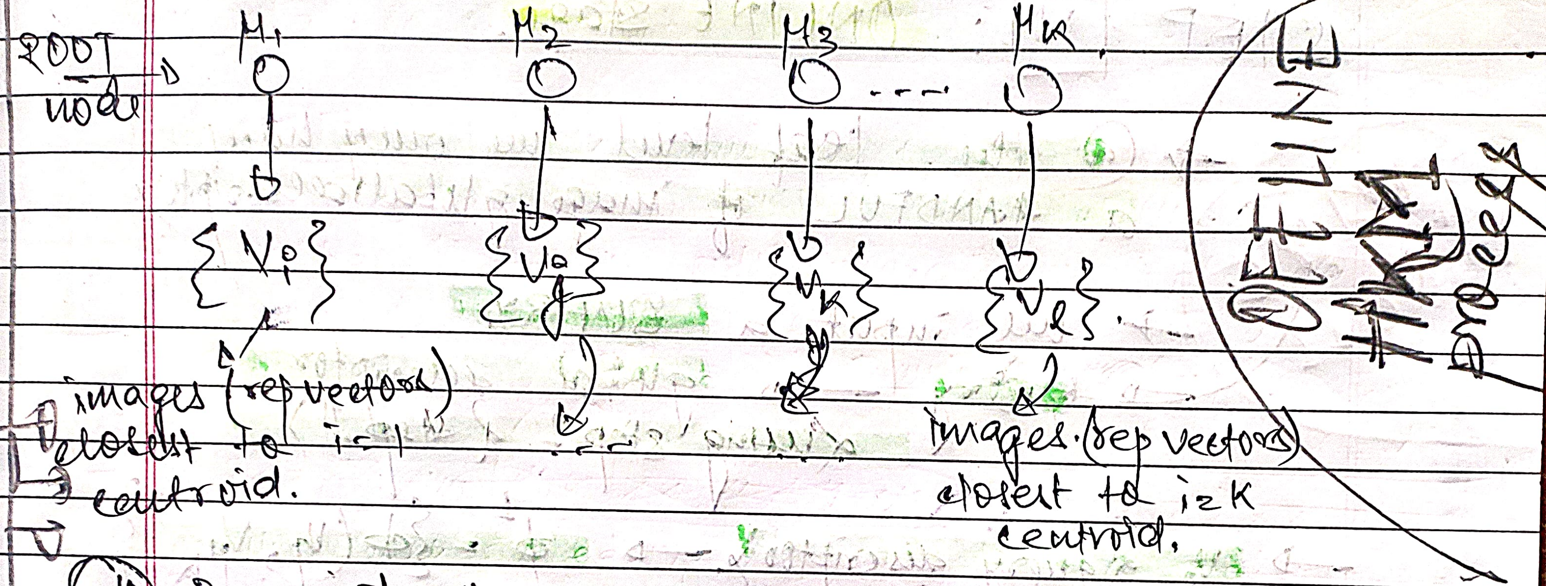

(*) Run iterative K means and correctly bucket images. After recursive K-means, we finally get the correct buckets (final K-buckets), each bucket has $\{b_i\}$ number of images.

$$\sum_{i=1}^{m} b_i^{(1)} = N$$

(*) For $l \geq 2$, we do K-means again, again get K buckets for each child.

The recursion is dependent upon "l" leaf/tree depth.

$$\sum_{j=1}^{m} b_{i,j}^{(2)} = b_i^{(1)} \Rightarrow \sum_{k=1}^{m} b_{i,j,k}^{(3)} = b_{i,j}^{(2)}$$

$$\Downarrow$$

$$\sum_{\alpha=1}^{m} b_{ijkl,\alpha}^{(5)} = b_{ijkl}^{(4)} \Leftarrow \sum_{l=1}^{m} b_{ijk,l}^{(4)} = b_{ijk}^{(3)}$$

(*) Each of these buckets at any depth WILL only contain A VERY SMALL subset of images.

(*) Leaf nodes → store ACTUAL image vectors
INTERNAL nodes → "label" → centroid vectors @ each level.

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STEP V: ONLINE Stage.

→ @ the leaf level we only have a HANDFUL of image global descriptors

→ we input a QUERY.

→ Query ⟶ global descriptor (using step I & step II).

→ V_q (query descriptor). → $\phi^1 = \{\phi(V_q, V_{M_K})\}$

just a sample!

choose argmin (ϕ^1).

choose argmin (ϕ^2) ⟶ move on to layer 2.

→ move over to layer 3 → choose argmin (ϕ^3).

→ TIP Now the similarity metrics were b/w V_q & $V_{M_K}^{(i)}$. Now that we have THE best Node. we choose THE candidate images in THAT node & calculate similarity vector.

→ suppose BEST node has b images, we calculate pairwise similarity ⟶ $\{\phi_1, \phi_2, \phi_3 \cdots \phi_b\}$.

AND then SORT these images.

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→ sorted $\{s_{q_1}, s_{q_2}, s_{q_3} \ldots s_{q_\beta}\}$ ⇒ RANKING.

THAT'S IT. Now, $\text{argmax}[\text{sorted}\{s_{q_i}\}] = \text{ANS}$

answer